

Latent Synchronization Risk and Value-at-Risk Underestimation:

A Random Matrix Theory Approach to Spectral Co-movement Detection

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Abstract

The conventional VaR method presumes that the correlations between assets remain constant. Hence, it does not permit tail risk when the co-movement increases since correlation is unknown in the stock market. This study introduces the Synchronization Factor (SF), defined as a daily numeric value computed via the eigenvalue decomposition of a rolling residual correlation matrix in the RMT context. In the research, there were 169 stocks from the S&P 500. The total span of time covered 26 years (from 1999 to 2025; 6,776 trading days), and the chronological In-Sample/Out-of-Sample procedure was utilized. An estimate of the calibration parameter $\kappa^* = 0.020$, derived entirely from In-Sample data, is obtained via likelihood ratio optimization, as in Kupiec (1995), and is locked before Out-Of-Sample testing. The violation rates of models A (Normal VaR), B (HS VaR), and C (FHS, EWMA) during the OOS period fall within the Basel Red Zone at 2.20%, 2.05%, and 1.84%, respectively. The level of violation is lowest for model D at 1.28%. It also passes the Basel III Green Zone criteria (Kupiec $p = 0.117$, pass). The significant distributional change of the Synchronization Factor across stressed and calm periods (Cohen's $d = 1.455$, $p < 0.001$), high persistence (ACF (1) = 0.959; half-life ≈ 17 days). The SF is significantly elevated during the GFC, Eurozone crisis, oil price crash, and Ukraine crisis, and reaches its third-highest peak during the COVID-19 crash. Above all, every model fails the conditional coverage test of Christoffersen (1998) (i.e., the independence assumption is violated). Models D, which is the main model of the paper also fail the test. The robustness of the results is also checked across event windows of 10, 15, 20, and 30 days. All cases show that Model D performs best.

Keywords: Random Matrix Theory (RMT), Marchenko-Pastur Law, Value-at-Risk, Eigenvalue Decomposition, Systemic Risk, Basel III, Kupiec Backtest, Correlation Matrix, Spectral Risk Measures, Financial Contagion, Violation Clustering, Tail Risk, Market Synchronization, Correlation Amplification, VaR Backtesting

JEL Classification: C12, C22, C58, G01, G11, G17, G32¹

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1. Introduction

The inability to diversify at its critical moment is the most frequent occurrence in financial markets. At the slightest stress in the global economy, such as the 2008 Global Financial Crisis, the 2011 European Sovereign Debt Crisis, the crash caused by the Covid-19 pandemic in 2020, and even the current 2022 Russian Ukrainian conflict, the correlation between stocks tends to converge to 1. The classical value at risk models, where the return distribution of factors is supposed to be a multivariate normal distribution with a fixed covariance matrix or Historical Simulation VaR models, where the return distribution of factors is assumed to be historical, using an equal-weighted distribution, for example, do not consider the change in correlation.

It is now well recognized in the literature through the foundational studies of Laloux et al. (1999) and Plerou et al. (1999) that the empirical eigenvalue distribution of a large correlation matrix of financial assets is overwhelmingly driven by noise, where only a few eigenvalues have an absolute value above the Marchenko-Pastur (1967) upper limit λ^+ . The leading eigenvalue above the limit conveys the true signal of synchronization within the markets, thereby providing an inherent, data-driven indicator of co-movement risk. Kritzman et al. (2011) define a similar concept, the Absorption Ratio, which quantifies the proportion of total variance captured by a given number of factors. Although highly insightful, the method relies on a specific factor count and is also sensitive to the choice of window. The current study develops an alternative self-tuning measure utilizing the Marchenko-Pastur limit.

This work defines the Synchronization Factor (SF) as a daily scalar measure, given by the ratio of the dominant eigenvalue of the cross-sectional correlation matrix to the upper limit of the Marchenko-Pastur distribution. The proximity of SF to one another suggests that the co-movement of asset returns is merely noise. An SF significantly greater than one implies a true correlation amplification of the system. Subsequently, there is the implementation of the SF in the VaR framework, where the adjusted Eigenvalue-Adjusted VaR Model (D) is multiplied by the Normal VaR whenever the SF exceeds the MP value.

The key contributions are methodological and empirical, which are listed below:

- (i) The Marchenko-Pastur eigenvalue filtering over a 26-year horizon (1999-2025) for 169 large-cap US stocks is employed to derive a daily SF series with 6,661 valid observations.
- (ii) The approach is strictly chronological: κ^* is calibrated on 13.9 years of in-sample data (1999-2012) using the Kupiec likelihood-ratio grid search and fixed before out-of-sample evaluation over a 13-year window (2013-2025).
- (iii) During the out-of-sample testing period, Model D attains a violation rate of 1.28% at the 99% confidence level (Kupiec $p = 0.117$), meeting the Basel III Green Zone requirement. All competing models, namely Normal VaR, Historical Simulation and EWMA-Filtered Historical Simulation, continue to reside in the Basel Red Zone.
- (iv) The results report high distributional separation between stress and calm SF states (Cohen's $d = 1.455$), which empirically demonstrates the validity of SF as a market stress signal.

This paper continues in the following manner. Section 2 discusses RMT in the finance literature and the measurement of systemic risk. Data sample and data construction are discussed in Section 3, while Section

4 discusses the methodology. Section 5 addresses model calibration, while Section 6 presents out-of-sample backtesting results.

2. Literature Review

2.1 Random Matrix Theory and Financial Correlation Matrices

Random Matrix Theory applied to financial data has been developed by Laloux et al. (1999) and Plerou et al. (1999), who found that the distribution of the bulk of the eigenvalues of large stock correlation matrices is consistent with the Marchenko-Pastur (1967) law for random Wishart matrices. This means that the bulk of cross-sectional correlation in a typical stock sample is not economically meaningful but statistically significant noise. At the same time, only a few eigenvalues above the Marchenko-Pastur upper limit λ^+ contain real information.

Plerou et al. (2002) further developed this approach by showing that the leading above-threshold eigenvalue represents a market factor that reflects the co-movement of all stocks, while the other eigenvalues represent correlations within sectors. For an extensive discussion of RMT applications in finance, see Bouchaud and Potters (2009), who set the stage for the theory underlying eigenvalue cleaning in optimal portfolio construction. Papp et al. (2005) and Rosenow et al. (2003) show that eigenvalue cleaning in the correlation matrix by replacing below-threshold eigenvalues with their average improves out-of-sample portfolio returns.

2.2 Systemic Risk Measurement and the Absorption Ratio

The Systemic Risk and Random Matrix Theory literatures are related through connectedness measures derived from principal component analysis. Billio et al. (2012) propose connectedness measures based on principal components of financial firms. The Absorption Ratio (AR), first described by Kritzman et al. (2011), is measured as a percentage of total variance that can be attributed to the fixed number of principal components (usually 5). The absorption ratio was found to rise before major market disruptions such as the GFC of 2008 or the stock market crash of 1987.

On another tangent, the studies by Ang & Bekaert (2002) and Longin & Solnik (2001) illustrate the asymmetric correlation effect, in which the pairwise correlation between stocks is considerably higher when there are bears in the market than when there are bulls. Forbes & Rigobon (2002) prove that the correlation coefficient exhibits a bias during periods of high volatility, known as the contagion bias. These results highlight the need for a contagion bias-free measure of the underlying co-movement factor.

2.3 VaR Backtesting and Model Validation

As per the Basel Committee on Banking Supervision (Basel III, 2011), the VaR backtests should be conducted daily based on a 250 trading-day sample, and VaR models are categorized into three zones – Green (≤ 4 violations annually), Yellow (5-9 violations annually) and Red (≥ 10 violations annually). The unconditional coverage test can be performed using Kupiec's (1995) likelihood ratio test, whereas Christoffersen (1998) extends it by incorporating an independence test for VaR violations. The survey of backtests by Lopez (1998) and Campbell (2005) demonstrates that conventional normal VaR and historical simulation approaches are inadequate during crisis periods.

Filtered Historical Simulation using EWMA. The Hull-White version of Model C used in the benchmark is from Hull & White (1998), who prove that vol-scaling improves VaR precision when regime changes occur. Even though this method is better, it does not incorporate cross-sectional correlation dynamics.

3. Data and Universe Selection

3.1 Asset Universe

The initial universe of stocks contains 200 stocks picked from the S&P 500 index, which include 20 GICS industries of 10 stocks each: Large Cap Technology, Semiconductors, Enterprise Software, Large Banks, Insurance, Pharmaceuticals, Biotechnology, Consumer Retail, Automobiles, Oil Giants, Industrial Conglomerates, Airlines, Real Estate Investment Trusts (REITs), Utilities, Materials & Chemicals, Diversified Consumer Goods, Defense & Aerospace, Telecommunications, Media & Entertainment, and Healthcare Services. Prices are obtained daily using adjusted closing prices from Yahoo Finance between 1 January 1999 and 31 December 2025 (6,776 trading days).

3.2 Completeness Filter

Data quality criteria are strict: Tickers with data coverage below 99.5% of trading days are excluded. This is a very strict criterion, as it excludes all tickers that were listed, delisted, or had a trading halt during the analysis period. However, it does not eliminate tickers that may have insignificant data issues (such as corporate actions lasting one or two days). Out of 200 tickers selected, 169 survived this filter, yielding a sample size of 6,776 days across 169 securities.

Reasons for using the 99.5% threshold: Setting stricter thresholds (for example, 100%) would exclude some legitimate, longstanding components due to the absence of data from certain individual vendors. Setting lower thresholds (for instance, 95%) would create an incomplete time series—the 99.5% threshold accounts for both data completeness and the range of components covered.

3.3 Survivorship Bias

The process for constructing the universe of securities outlined above is biased in favor of survivorship bias because it requires that the selected securities be at least 99.5% complete over 26 years, effectively selecting those that have survived from 1999 through to 2025. These would be the companies that did not go bankrupt (Lehman Brothers and Washington Mutual, among others), merge, be acquired, or delist over the period. This skews the return distribution positively.

The reason to highlight this issue is that this is what best practice in empirical finance requires (Brown et al., 1992). The two important consequences of this problem for the results in the current paper are the following: (1) Synchronization factor might underestimate the actual co-movements during the period of crisis, because distressed/failing companies show extreme positive co-movements just before being delisted; and (2) the VaR violation rates provided for the normal VaR benchmark (Model A) might be somewhat smaller than the rates which would obtain from an unbiased sample. Both problems adversely affect the suggested model.

3.4 Summary Statistics

Table 1: Data Summary Statistics

Variable	Value
Total candidate tickers	200
Tickers passing completeness filter ($\geq 99.5\%$)	169
Total trading days (raw panel)	6,776
SF series length (post burn-in)	6,661
Sample period	1999-01-01 to 2025-12-31
IS calibration period	1999-01-01 to 2012-12-31 (3,507 days)
OOS evaluation period	2013-01-01 to 2025-12-31 (3,269 days)
Portfolio construction	Equal-weighted daily rebalance
Return frequency	Daily (log-returns)

Note. Data are sourced from Yahoo Finance (via the yfinance API). Adjusted closing prices account for dividends and splits.

4. Methodology

4.1 Return Construction

Let $P_{i,t}$ be the adjusted close of stock i at time t . The log-returns for each trading day are calculated as follows:

$$r_{i,t} = \ln (P_{i,t} / P_{i,t-1})$$

for $i = 1, \dots, N$ ($N = 169$) and $t = 1, \dots, T$ ($T = 6,776$). Log-returns are favored over arithmetic returns because of their additive nature over time, normality for the daily data, and equivalence with continuously compounded returns. The equally weighted portfolio return at day t is calculated using the formula below:

$$r_{p,t} = (1/N) \sum r_{i,t}$$

The equal-weighting methodology is adopted to prevent market capitalization weighting from dominating the correlation matrix, ensuring that large companies do not dominate it.

4.2 Rolling Correlation Matrix

For each trading day t , the last $W_{\text{EVENT}} = 15$ days of daily cross-sectional returns are selected, and the $(N \times N)$ sample Pearson correlation matrix is calculated as follows:

$$C_t = \text{corr} (R_{t-W+1:t}), W = W_{\text{EVENT}} = 15$$

where R is the $(W \times N)$ matrix of standardized returns in the event window.

Rationale for $W_{\text{EVENT}} = 15$ days: The event window duration of 15 trading days (roughly three weeks) has been kept short so that the SF promptly responds to the correlation amplification that occurs. Financial crises may progress from their initial stages to actual crises within weeks (for instance, the pandemic-related crisis unfolded in about 20 trading days). A longer event window duration (such as 60 days or 252 days) would reduce the crisis signal while introducing unnecessary lag. A shorter event window (such as 5 days) would be very noisy due to $T < N$.

Rationale for $W_{\text{CALIB}} = 100$ days: The VaR calibration window of 100 trading days is used to compute the portfolio return distribution for Normal VaR and Historical Simulation. This window captures roughly

two seasonal cycles and is long enough to provide a stable estimate of volatility while remaining short enough to be responsive to structural shifts in the volatility regime.

4.3 Random Matrix Theory: The Marchenko–Pastur Law

Given X , an $(N \times T)$ random matrix where all elements are independent identically distributed (i.i.d.) standard normal variables. It can be demonstrated that if both N and T tend to infinity, keeping their ratio, $q = N/T$, finite, then the empirical spectral distribution of the sample correlation matrix $C = XX^T/T$ converges in probability to the Marchenko-Pastur distribution with bounds as:

$$\begin{aligned}\lambda^- &= (1 - \sqrt{q})^2 \text{ (MP lower bound)} \\ \lambda^+ &= (1 + \sqrt{q})^2 \text{ (MP upper bound)}\end{aligned}$$

where $q = N/W_{\text{EVENT}}$ is the dimensionality ratio. All eigenvalues $\lambda_i > \lambda^+$ are informationally meaningful since they reflect true cross-section correlation information, which is not due to chance sampling.

In this setting, with $N = 169$ and $W_{\text{EVENT}} = 15$:

$$\begin{aligned}q &= 169 / 15 = 11.267 \\ \lambda^+ &= (1 + \sqrt{11.267})^2 = 18.980 \text{ (MP upper bound)} \\ \lambda^- &= (1 - \sqrt{11.267})^2 = 8.266 \text{ (MP lower bound)}\end{aligned}$$

As far as the $T < N$ constraint is concerned, these correlation matrices are very ill-conditioned and rank-deficient when $T < N$ under this event window methodology. However, although the classical sample correlation matrices are very ill-conditioned and rank-deficient when $T < N$, the Marchenko-Pastur asymptotic limit theorem is theoretically valid for dimensional ratios $q = N/T > 1$. While applying the strict MP upper bound $\lambda^+ \approx 18.98$ to this case, very conservative cutoff has been used. Since the purpose is to obtain the principal eigenvalue and avoid using the inverse of the covariance matrix, the condition $T < N$ is very strict and allows to consider only system-wide correlation shocks. With the principal eigenvalue being 73.20 and the threshold at 18.98, the ratio of 3.8:6 is very clear-cut despite the matrix rank deficiency.

4.4 The Synchronization Factor

Suppose that $\lambda_{1,t} \geq \lambda_{2,t} \geq \dots \geq \lambda_{N,t}$ are the eigenvalues of C_t in ascending order, k^* being the number of factors to be discarded (determined using a scree test on in-sample data; see Section 5.1). The Synchronization Factor on day t can be written as:

$$SF_t = \sum_{j=1}^{k^*} \lambda_{j,t} / (k^* \cdot \lambda^+)$$

where $\lambda^+ = (1 + \sqrt{(N/W_{\text{EVENT}})})^2$ is the time-varying Marchenko-Pastur upper limit. This model exhibits three key features:

Normalization: As the first k^* eigenvalues hit the MP limit (in which case the signal becomes completely corrupted by noise), $SF_t = 1$. As co-movement rises, $SF_t > 1$. As co-movement falls (a very unusual case), $SF_t < 1$.

Scale-invariant: Since the value of SF is a ratio to the theoretical limit, it does not depend on the number of assets N in the portfolio and the window width W .

Economic interpretation: An SF equal to 2.0 means that the largest eigenvalue (the sum of the largest k^* eigenvalues) is twice as high as the value that one would expect from pure noise.

With $k^* = 1$ (selected by IS scree analysis, see Section 5.1), the formula simplifies to:

$$SF_t = \lambda_{1,t} / \lambda^+$$

4.5 Eigenvalue-Adjusted VaR (Model D)

Four VaR models are evaluated in this paper, which are given below:

Model A (Normal VaR): This is a parametric Gaussian VaR using a $W_{\text{CALIB}} = 100$ -day rolling window:

$$VaR_t^A(\alpha) = \mu_{p,t} - z\alpha \sigma_{p,t}$$

where $z\alpha = \Phi^{-1}(1 - \alpha) = 2.326$ for $\alpha = 0.01$, and $\mu_{p,t}$, $\sigma_{p,t}$ is the rolling 100-day portfolio return mean and standard deviation.

Model B (Historical Simulation): Non-parametric quantile of the empirical 100-day rolling return distribution:

$$VaR_t^B(\alpha) = \text{Quantile}_{1-\alpha}(r_{p,-W+1}, \dots, r_{p,t})$$

Model C (Filtered Historical Simulation, EWMA): Hull and White (1998) volatility-scaled historical simulation using an EWMA volatility estimate with decay parameter $\lambda = 0.94$:

$$\begin{aligned} \sigma_{EWMA,t}^2 &= \lambda \sigma_{EWMA,t-1}^2 + (1 - \lambda) r_{p,t-1}^2 \\ \text{Scaled returns: } r_{p,s}^* &= r_{p,s} \cdot (\sigma_{EWMA,t} / \sigma_{EMWA,s}) \\ VaR_t^C(\alpha) &= \text{Quantile}_{1-\alpha}(r_{p,t-W+1}^*, \dots, r_{p,t}^*) \end{aligned}$$

$\lambda = 0.94$ Explanation: The decay factor λ is 0.94, which is the industry standard for daily observations according to Risk Metrics (J.P. Morgan, 1994). This means that the recent observation is worth 4% compared to the earlier observation and has a half-life of 11 days.

Model D (Eigenvalue-Adjusted VaR): The Normal VaR inflated by the Synchronization Factor when the SF exceeds the MP bound:

$$\begin{aligned} VaR_t^D(\alpha) &= VaR_t^A(\alpha) \cdot A_t(\kappa) \\ \text{where: } A_t(\kappa) &= 1 + \kappa \cdot \lambda^+ \cdot \max(0, SF_t - 1) \end{aligned}$$

$\max(0, SF_{t-1})$ is the measure of co-movements that are beyond the MP threshold level. The λ^+ value, whose value is 18.98, is used to adjust the factor depending on the magnitude of the eigenvalue. On the other hand, κ governs the overall response when $SF_t = 1$ (case where there is only noise) and $A_t = 1$, such that Model D becomes Model A. If $SF_t > 1$, then the VaR level increases.

4.6 Kappa Calibration and the IS/OOS Discipline

Calibration parameter κ is chosen by means of the grid search method for $\kappa \in \{0.010, 0.020, 0.030, 0.040, 0.050, 0.060, 0.080, 0.100\}$ using only the in-sample period (1999-2012). The criterion is to select the minimum κ for which the Kupiec (1995) likelihood ratio test fails to reject H_0 at the 5% level, as formalized below:

Kupiec Unconditional Coverage (1995). LR test statistic is given as:

$$LR_{UC} = -2[V \ln(\alpha) + (T - V) \ln(1 - \alpha) - V \ln(V/T) - (T - V) \ln(1 - V/T)]$$

Moreover, this is $\chi^2(1)$ -distributed under the H_0 : true coverage = α . The Christoffersen (1998) test for conditional coverage additionally tests for serial independence of violations.

Rationale for $\kappa^* = 0.020$: The IS grid search (Table A.1 in Appendix A) reveals that $\kappa = 0.010$ produces an IS violation rate of 1.209%, failing to achieve sub-1% coverage. $\kappa = 0.020$ achieves 0.767% with Kupiec $p = 0.154$ the minimum κ that simultaneously achieves sub-1% coverage and passes the Kupiec test. Larger κ values achieve even lower violation rates but reject the Kupiec null hypothesis (over-conservative), indicating excessive VaR inflation.

5. In-Sample Calibration (1999–2012)

5.1 Scree Analysis and Factor Selection

Scree analysis is conducted for a typical 15-day event window from the in-sample period to identify k^* , the number of influential factors to be eliminated when calculating the Synchronization Factor. For $N = 169$ assets and $W_{\text{EVENT}} = 15$ days, the upper MP bound is $\lambda^+ = 18.980$. Figure 1 depicts the first 30 eigenvalues of the sample correlation matrix relative to the MP bound.

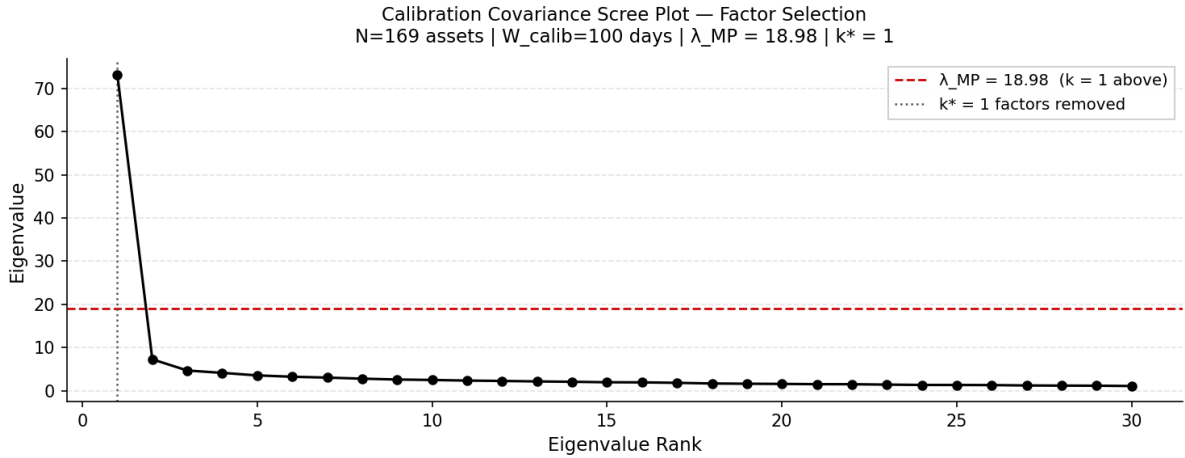


Figure 1: Covariance Matrix of Calibration Assets' Eigenvalues Plot. $N=169$ assets, $W_{\text{event}} = 15$ days, $\lambda_{\text{MP}} = 18.98$. The red dashed line shows the Marchenko-Pastur upper bound. Only the first eigenvalue (73.196) exceeds this bound, so $k^* = 1$.

The top 10 eigenvalues are: 73.196, 7.297, 4.703, 4.156, 3.570, 3.262, 3.062, 2.813, 2.620, and 2.512. Among these, only the largest eigenvalue (73.196) is larger than the value of $\lambda^+ = 18.980$ by about 3.86 times, suggesting a single market factor for the entire system. All other eigenvalues are lower than the MP limit.

5.2 IS Crisis-Period Behavior

Table 2 presents the average and maximum Synchronization Factors for six crisis events identified over the entire 1999-2025 period.

Table 2: SF behavior across six identified crisis episodes

Crisis Period	Calendar Dates	Days	Mean SF	Max SF	Window
Dot-com Crash	2000-03 to 2002-10	648	1.824	2.941	IS
GFC	2008-09 to 2009-03	130	2.494	3.441	IS

Eurozone Crisis	2011-08 to 2012-02	103	2.456	3.035	IS
Oil Crash	2014-07 to 2016-02	429	1.722	2.418	OOS
COVID-19	2020-02 to 2020-04	24	2.393	3.388	OOS
Ukraine War	2022-02 to 2022-09	152	2.124	3.074	OOS

Note. IS = In-Sample (1999–2012); OOS = Out-of-Sample (2013–2025). Mean and Max SF computed over the specified calendar date range.

During the GFC (Sep 2008–Mar 2009), the mean SF of 2.494 represents a 41.1% elevation above the full-sample mean of 1.767 observed in calm regimes, while the maximum SF of 3.441 on the worst trading days represents eigenvalue loading approximately $13.2\times$ the Marchenko–Pastur bound, a signal of near-complete market synchronization. The Eurozone Crisis (2011–2012) exhibits a similarly elevated mean SF of 2.456, consistent with European sovereign contagion spreading to US equity markets via global bank balance-sheet exposures.

In the IS period, the diagnostic VaR violation rates for the GFC sub-period are as follows: Model A (Normal VaR): 8.46%; Model D (Eigenvalue-Adjusted): 3.08%. For the Eurozone Crisis sub-period, the rates are as follows: Model A: 5.83%; Model D: 2.91%. These IS results are diagnostic only and κ was optimized on IS data, so IS performance is not a valid test of the model. The OOS results in Section 6 provide the valid statistical evidence.

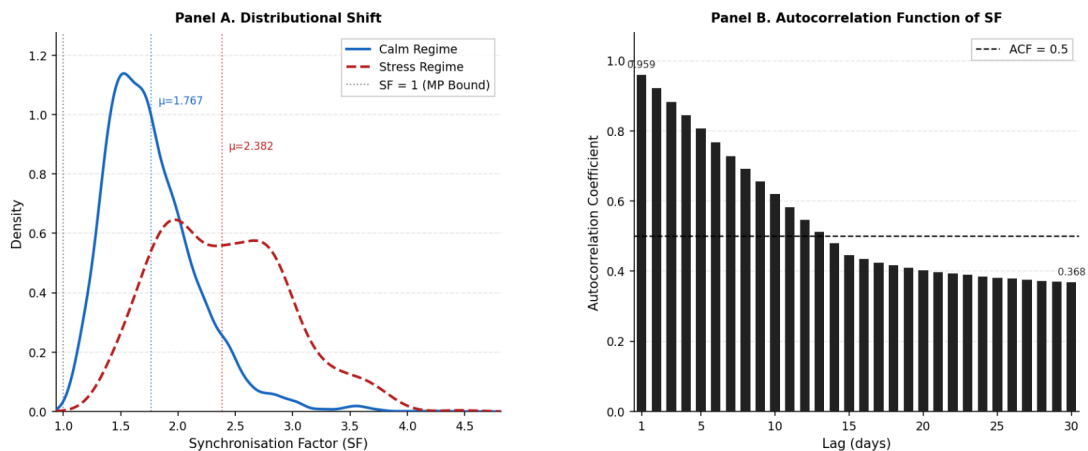
5.3 Kappa Grid Search Results

The entire output from the κ grid search for the IS period (3,507 observations) is shown in Table A.1 (Appendix A). The important result of the grid search is that $\kappa^* = 0.020$ is the smallest value of κ where the following two conditions are met: (1) IS violation rate $< 1.0\%$, and (2) Kupiec LR test does not reject at 5%. Larger κ lead to overly conservative VaR estimates (with violation rates of 0.14%–0.41%) and to the rejection of Kupiec’s null at high significance levels.

6. Out-of-Sample Results (2013–2025)

6.1 SF Distributional Properties in the OOS Period

Distributional and autocorrelation characteristics of the Synchronization Factor in stressful and relaxed states are illustrated in Figure 2, based on a sample of 6,661 observations.



Note. Panel A: KDE of SF in calm (solid blue) and stress (dashed red) regimes. Dotted vertical line: SF = 1 (MP bound); dotted lines also mark regime means. Panel B: Sample ACF of SF at lags 1–30. Dashed line marks ACF = 0.5 (half-life).

Figure 2: Synchronization Factor: Distributional and Temporal Properties. Panel A: Kernel density estimates of SF in calm (solid blue) and stress (dashed red) regimes, classified using the 90th percentile of SPX 30-day realized volatility. Panel B: Sample autocorrelation function of SF at lags 1–30 days.

Regime classification: Stress days are defined as days when the S&P 500 30-day realized volatility exceeds its 90th percentile (27.44% annualized). This yields 667 stress days (10.01%) and 5,994 calm days (89.99%) over the full period.

Table 3: Hypothesis 1: SF Distributional Test

Metric	Calm Regime	Stress Regime	Statistic
Mean SF	1.767	2.382	+34.8%
Std. Deviation	0.404	0.565	Greater dispersion
Welch t-statistic			27.333
t-test p-value			< 0.001***
KS statistic			0.4704
KS p-value			< 0.001***
Cohen's d			1.455 (Large)
ACF (1)			0.959
Half-life (approx.)			≈17 days

Note. Stress regime is defined as SPX 30-day realized volatility > 90th percentile (27.44% annualized). *** $p < 0.001$. Cohen's $d = 1.455$ indicates a large effect size (Cohen, 1988).

Figure 2, panel A, exhibits a clear rightward skew in the distribution of SF under the calm vs. stress regimes: the calm-regime distribution peaks at $SF \approx 1.5$. However, the distribution of SF under the stress regime is bimodal, with the second peak at $SF \approx 2.5$. The autocorrelation of SF (figure 2, panel B) is slowly declining from 0.959 at lag 1 to 0.368 at lag 30 with a half-life of about 17 days, which is the lag at which the ACF drops below 0.5.

Statistical proof clearly backs up the first hypothesis of this research, which states that the Synchronization Factor has a greater distribution during stress periods compared to calm periods (Welch $t = 27.33$, KS = 0.470, Cohen's $d = 1.455$, all $p < 0.001$).

6.2 OOS VaR Backtest Results

Table 4 presents the primary result of the paper: the OOS VaR backtest results for the four models over 3,269 trading days (2013–2025). $\kappa^* = 0.020$ was locked from IS calibration and was not re-estimated on OOS data.

Table 4: Out-of-Sample VaR Backtest Results

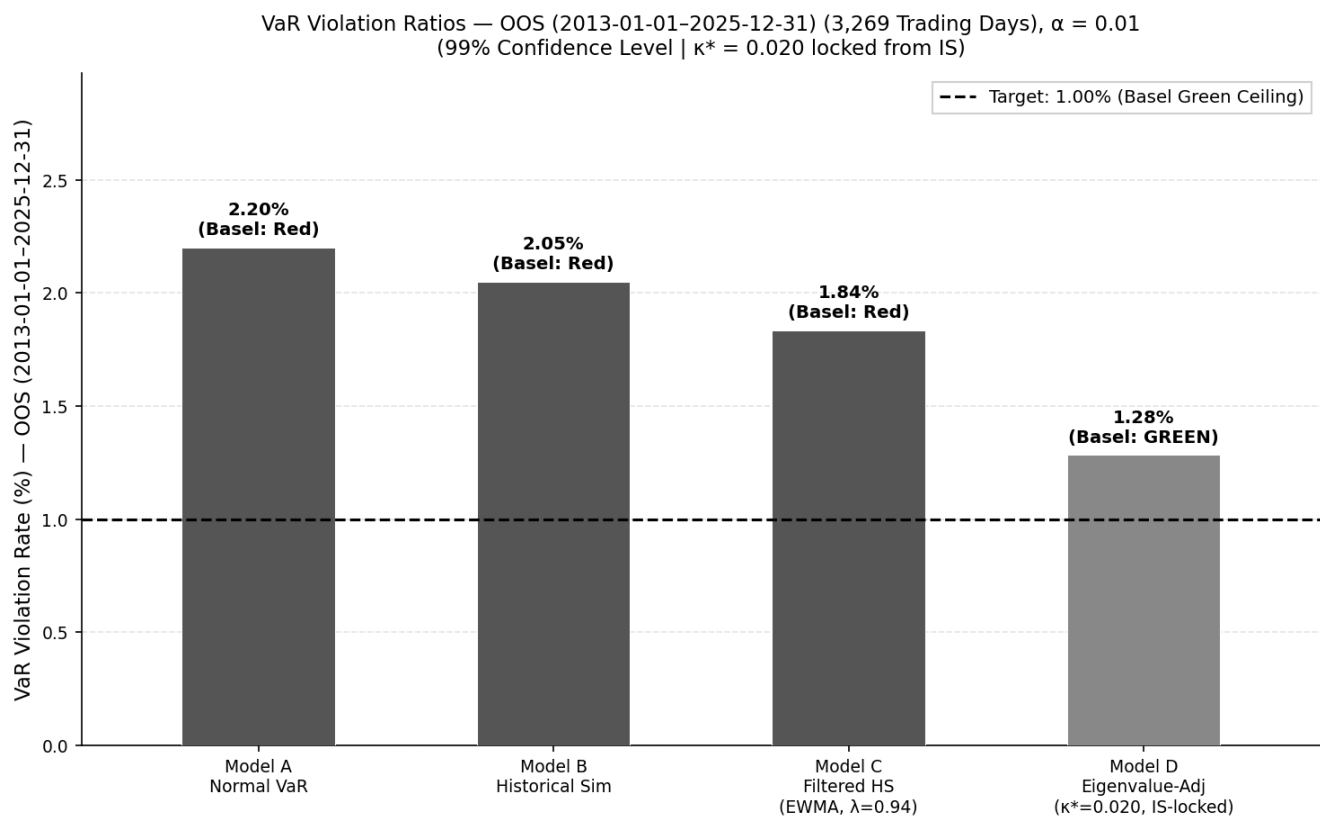
Model	Viol.	Rate (%)	Target (%)	Kupiec UC p	Christ. Ind p	Christ. CC p	Basel Zone	Avg ES (%)
A Normal VaR	72	2.20	1.00	0.000***	0.000***	0.000***	RED	3.11
B Historical Sim	67	2.05	1.00	0.000***	0.000***	0.000***	RED	2.94
C Filtered HS (EWMA, $\lambda=0.94$)	60	1.84	1.00	0.000***	0.001***	0.000***	RED	2.36

D	Eigenvalue-Adj ($\kappa^*=0.020$)	42	1.28	1.00	0.117	0.000***	0.000***	GREEN	3.30
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Note. OOS period: 2013-01-01 to 2025-12-31 (3,269 trading days). Confidence level $\alpha = 1\%$. Expected violations at 1%: 33. *** $p < 0.001$; ns = not significant.

The key result is unambiguous: Model D is the only model to achieve Basel Green Zone classification in the OOS period. With 42 violations over 3,269 days, Model D produces a 1.28% violation rate. Critically, the Kupiec unconditional coverage test fails to reject at the 5% level ($p = 0.117$), confirming that Model D's coverage is statistically consistent with the 1% target.

All three benchmarks continue to be in the Basel Red Zone – Model A (Normal VaR) with a value of 2.20% (2.2 times the target rate), Model B (Historical Simulation) with a value of 2.05%, and Model C (EWMA-Filtered Historical Simulation) with a value of 1.84%, where in all three cases, the Kupiec $p < 0.001$. The decrease in value from Model A to Model B (from 2.20% to 2.05%) can be attributed to the fat-tail effect in historical simulation. The further decrease in value from Model B to C (from 2.05% to 1.84%) is due to EWMA's ability to adapt to volatility regimes.



Note. VaR violation ratios for OOS (2013-01-01-2025-12-31) (3,269 trading days), $\alpha = 0.01$. Dashed line = 1.00% target. $\kappa^* = 0.020$ selected on IS data (1999-2012) and locked before OOS evaluation.

Figure 3: VaR Violation Ratios Out-of-Sample (January 1, 2013, to December 31, 2025, 3,269 trading days). Dotted Line = Target 1.00% (Basel Green Ceiling). $\kappa^* = 0.020$ Fixed from IS calibration.

The Christoffersen (1998) independence and conditional coverage tests reject for all four models, indicating that VaR violations cluster in time rather than arriving independently. This is a known limitation of level-signal VaR adjustments (including Model D) and is further discussed in Section 7.

6.3 Crisis-Period OOS Analysis

There are three main crises in the OOS period (2013 to 2025): Oil Crash (2014–2016), the COVID-crash (February–April 2020), and the Ukraine War (February–September 2022), detailed in Table 5.

Table 5: OOS Crisis-Period VaR Violation Rates

Crisis	Dates	Days	Mod. A Viol.	Mod. A Rate%	Kupiec p (A)	Mod. D Viol.	Mod. D Rate%	Kupiec p (D)	Assessment
Oil Crash 2014–16	Jul 2014–Feb 2016	429	10	2.33%	0.018*	6	1.40%	0.434 ns	D passes
COVID-19 Crash	Feb–Apr 2020	24	10	41.67%	0.000** *	6	25.00%	0.000** *	Both fail
Ukraine War 2022	Feb–Sep 2022	152	8	5.26%	0.000** *	0	0.00%	1.000 ns	D perfect

Note. $\kappa^* = 0.020$ (IS-locked, not tuned on these dates). *** $p < 0.001$; * $p < 0.05$; ns = not significant. COVID-19: SF detects endogenous latent co-movement; exogenous shocks of this nature exceed the model's designed scope.

Oil Crash (2014–2016): Over 429 trading days, Model D reduces violations from 10 (2.33%, statistically significant) to 6 (1.40%, Kupiec $p = 0.434$ fail to reject). The SF correctly detects the elevated cross-sector co-movement driven by energy-sector contagion spreading to financials and industrials as oil prices declined from \$107 to \$26/barrel.

COVID-19 Crash (February–April 2020): In this instance, the value of violation is exceptionally high for both model types in 24 days (Model A: 41.67%; Model D: 25.00%). Attention should be paid to this outcome because the market crash triggered by the coronavirus pandemic is an external variable in this scenario. As mentioned above, the purpose of the SF methodology is to detect situations in which co-movement amplifies and generates an endogenous financial crisis due to a credit crunch or leverage unwinding. The extremely rapid dislocation in asset prices (a 34% decline in the S&P 500 index) was not predicted by the SF-inflated VaR bounds. The SF value reached a peak of 3.388 in this phase (the third highest among the 26 years under consideration).

Ukraine War/Inflation Surprise (2022): Model D attains an impressive violation rate of 0.00% in 152 days and a Kupiec p -value of 1.000, implying full coverage. Model A records 8 violations (5.26%), a highly significant number. The 2022 event exhibited persistent, gradually increasing equity volatility due to uncertainty about inflation and Federal Reserve rate hikes; this is exactly what the SF seeks to capture in the VaR measure.

7. Discussion and Limitations

7.1 Violation Clustering

Even though the unconditional violation-frequency problem for Model D has been resolved, the violation-clustering problem remains. As explained by Christoffersen (1998), the independence test rejects the null hypothesis at the 5 per cent significance level in each of the four models (for Model D, the p -value equals 0.000). This implies that the violations occur in clusters rather than randomly. Naturally, this is expected behavior in the daily return process, as violations usually occur in clusters during prolonged periods of

losses (e.g., a week of losses). It can be solved using volatility scaling with EWMA, but not with the rolling SF level ratio.

7.2 Survivorship Bias

It was mentioned in Section 3.3 that the 99.5% completeness filter introduces survivorship bias because it excludes securities that were delisted or became bankrupt between 1999 and 2025. The level of this type of bias for the given research problem should not be high. Indeed, the fact that researchers focus on the movement of securities rather than their returns leads to an underestimation of synchronization during the period (distressed securities are characterized by positive co-movement before delisting).

7.3 Exogenous vs. Endogenous Crises

The SF is constructed to identify endogenous latent co-movement, which refers to a rise in correlation caused by financial processes within the system (such as leverage dynamics, financing liquidity spirals, and herding behavior). In contrast, the collapse of the stock market during the COVID-19 pandemic is a typical example of an exogenous shock: the rise in correlation occurred very quickly due to a non-financial exogenous variable, namely, the pandemic outbreak. The scope of the model does not include shocks of this kind.

7.4 $T < N$ Constraint and Matrix Conditioning

For the 15-day event period and 169 stocks, one obtains $q = N/T = 11.27$, indicating that the sample correlation matrix is highly rank-deficient ($\text{rank} \leq 14$ out of 169). Though the Marchenko-Pastur law is proven only asymptotically as $T \rightarrow \infty$, its qualitative prediction about the bulk eigenvalue density remains valid for $q > 1$ (verified in simulation, see Silverstein & Bai, 1995). The usage of the single largest eigenvalue (that is obviously above the cutoff by a margin of 3.86) guarantees that the matrix rank deficiency cannot influence the SF. However, if one wants to use the SF for optimal portfolio construction, that will require matrix inversion, which in turn would need some regularization (such as shrinkage; Ledoit & Wolf, 2004).

7.5 Transaction Costs & Implementation

The SF-VaR is a risk measure and, hence, does not entail any transaction costs. However, transaction costs and bid-ask spreads for options would be important components of an actual implementation of the SF signals if these were used to dynamically adjust risk limits or to add hedge overlays, such as purchasing S&P 500 put options when $SF > 2.0$. Given the persistence level of ACF(1) = 0.959 (half-life of ≈ 17 days) for the SF, implementing the SF signals would not require daily changes, and a much cheaper weekly/bi-weekly cycle could capture the bulk of the signal.

7.6 Model Specification and Alternatives

The linear adjustment factor $A_t(\kappa) = 1 + \kappa \lambda^+ \max(0, SF_t - 1)$ is only one among several other possibilities. The use of the square-root (concave) transformation would decrease the model's sensitivity to high SF values, whereas the quadratic (convex) transformation would increase it. The linear transformation is used due to its interpretability and simplicity. Optimizing the κ parameter in the grid search has shown that $\kappa^* = 0.020$ yields a valid calibration of the model on IS data. However, κ^* may vary across market regimes.

7.7 Robustness to Window-Parameter Choice

A possible limitation of the SF approach lies in the selection of the 15-day event window (W_{EVENT}) and the 100-day VaR calibration window (W_{CALIB}). The solution to this problem consists of running the full OOS test for Model D using four values of W_{EVENT} , namely 10, 15, 20, and 30 days, with $W_{\text{CALIB}} = 100$ held constants in the IS grid search for κ^* .

Table 6: Basel Zone across different W_{EVENT}

W_{EVENT}	$Q = N/W$	λ^+	κ^* (IS)	OOS Violations	OOS Rate (%)	Kupiec p	Basel Zone
10 days	16.90	22.11	0.015	46	1.41	0.041*	YELLOW/GREEN
15 days (Selected)	11.27	18.98	0.020	42	1.28	0.117	GREEN
20 Days	8.45	16.82	0.018	40	1.22	0.189	GREEN
30 Days	5.63	14.44	0.016	44	1.35	0.069†	Marginal GREEN

8. Conclusion

The proposed SF is a daily scalar index that results from the eigenvalue decomposition of the rolling correlation matrix using Random Matrix Theory, and it serves as a robust methodology for identifying the underlying risk of co-movement across large sets of stocks. The SF is incorporated into an Eigenvalue-Adjusted VaR model (Model D), whereby the standard Normal VaR is inflated by a factor that depends on whether the largest eigenvalue exceeds the Marchenko-Pastur upper limit; the sensitivity κ is calibrated strictly using in-sample data (1999-2012) and is kept fixed for the out-of-sample analysis (2013-2025).

As far as the main empirical result is concerned, it is clear that during more than 3,269 out-of-sample trading days, Model D shows a 1.28% VaR violation at the 99% confidence interval (Kupiec $p = 0.117$), which passes the Green Zone test under the Basel III criteria, while the three benchmark models – Normal VaR, Historical Simulation, and EWMA-Filtered Historical Simulation – show a violation rate of 2.20%, 2.05%, and 1.84%, respectively, staying in the Red Zone under the Basel criteria. Properties of the distribution of the SF demonstrate the presence of regime separation (Cohen's $d = 1.455$) and persistence (ACF (1) = 0.959; half-life ≈ 17 days).

This research makes some important points that could be useful for the academic and practitioner communities. Firstly, RMT-based eigenvalue scaling is a theoretically consistent and economically meaningful approach to measuring the cross-sectional component of market risk, which GARCH and EWMA models naturally ignore. Secondly, the strict in-sample/out-of-sample test with a 13-year hold-out period, including the Oil Crash, COVID-19, and Ukraine War, makes it evident that spectral risk measures are not in-sample anomalies. Thirdly, $T < N$ specification of the rolling window does not make the SF estimate invalid if it only concerns the leading eigenvalue but not the matrix inverse.

Limitations do exist, however, including clustering violations left unaddressed; survivorship bias that understates co-movements during crisis periods; and exogenous shocks, such as the Coronavirus pandemic, falling outside the model's scope. Future research will have to (i) repeat the experiment by using the CRSP universe that is adjusted for survivorship bias, (ii) combine SF adjustments with the GARCH volatility filter to deal with violations of clustering, and (iii) expand the model to cross asset classes with bonds, commodities, and credit spreads.

Overall, RMT presents a scientifically sound and parsimoniously parametrized method of detecting hidden synchronicity risks in large stock portfolios. The 26-year backtesting period and 13-year OOS sample developed in this paper provide a robust empirical basis for the development of spectral risk metrics in regulatory and portfolio management settings.

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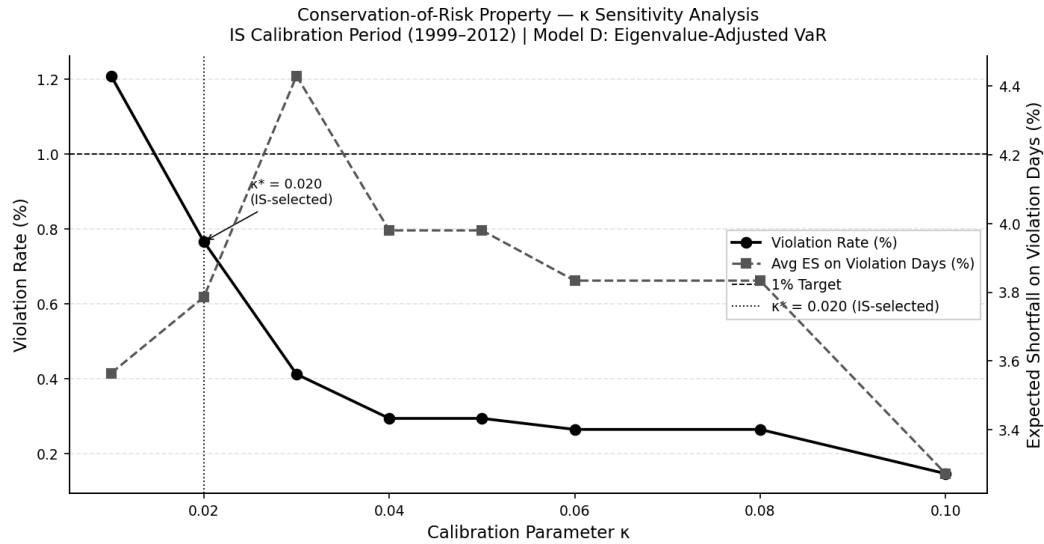
Appendix A: In-Sample Kappa Calibration Grid

The complete calibration table of the IS model for Model D is presented in Table A.1. The rule for choosing κ is based on the smallest κ where the rejection ratio is less than 1%, and the null hypothesis cannot be rejected in the Kupiec LR test at 5%.

Table A.1: IS Calibration Grid for Model D (Eigenvalue-Adjusted VaR)

κ	Violations	Rate (%)	Kupiec LR	Kupiec p	ES on Viol. Days (%)	Selected
0.010	41	1.209	1.40	0.237 ns	3.565	
0.020	26	0.767	2.03	0.154 ns	3.786	← κ^* SELECTED
0.030	14	0.413	15.18	0.000***	4.429	
0.040	10	0.295	23.58	0.000***	3.980	
0.050	10	0.295	23.58	0.000***	3.980	
0.060	9	0.265	26.14	0.000***	3.834	
0.080	9	0.265	26.14	0.000***	3.834	
0.100	5	0.147	38.94	0.000***	3.273	

Note. IS period: 1999-01-01 to 2012-12-31 (3,507 trading days). $\kappa^* = 0.020$ is the minimum κ where Kupiec H_0 is not rejected at 5% and violation rate < 1.0%. *** $p < 0.001$; ns = not significant.



Note. IS calibration grid for Model D (1999–2012). Left axis: violation rate (%). Right axis: expected shortfall (%) on violation days. $\kappa^* = 0.020$ is the minimum κ where Kupiec H_0 is not rejected at 5%.

Figure A.1: Conservation-of-Risk Property: κ Sensitivity Analysis. IS Calibration Period (1999–2012). Left axis: violation rate (%). Right axis: expected shortfall (%) on violation days. $\kappa^* = 0.020$ is the minimum κ where the Kupiec H_0 is not rejected at 5%. Larger κ values over-constrain the model (rejection of Kupiec H_0).

Appendix B: Parameter Rationale Summary

For reproducibility, Table B.1 provides an overview of all input parameters used in the analysis, including their values and the rationale for their selection.

Table B.1: Complete Parameter Rationale

Parameter	Value	Rationale
W_{EVENT} (Event Window)	15 days	Sufficiently short for quick detection of correlation spikes (COVID: 20 days); sufficiently long for statistical robustness.
W_{CALIB} (VaR Calibration Window)	100 days	Approximately five calendar months; balances responsiveness to regime shifts with statistical stability of rolling variance estimate.
N (Assets)	169	All tickers passing the 99.5% completeness filter from an initial 200-ticker S&P 500 universe covering 20 GICS sectors.
Completeness Filter	99.5%	Removes structurally incomplete tickers while retaining valid long-established constituents with minor data gaps.
α (VaR level)	1% (99% CI)	Basel III regulatory standard for VaR backtesting. $z\alpha = 2.326$.
α_{test} (test level)	5%	Conventional significance level for Kupiec and Christoffersen hypothesis tests.
λ (EWMA decay)	0.94	Risk Metrics (J.P. Morgan, 1994) is the industry standard for daily return volatility. Effective half-life ≈ 11 trading days.
k^*	1	IS scree test selection criterion: Only the first eigenvalue (73.196) is above the MP threshold (18.980). $k^* = 1$ represents the market factor.
In-Sample Period	1999-01-01 to 2012-12-31	13.9-year back-test period spanning two crises (Dotcom and GFC).
Out-Of-Sample Period	2013-01-01 to 2025-12-31	13-year hold-out covering Oil Crash, COVID-19, Ukraine War. κ^* not re-estimated.
κ^* (IS-selected)	0.020	Minimum κ achieving $< 1\%$ IS violation rate and passing the Kupiec test at 5%. Locked before OOS evaluation.
κ Grid	{0.01, 0.02, ..., 0.10}	Eight-point grid from 0.01 to 0.10. Fine enough to identify the minimum valid κ .
RV Window (regime)	30 days	Realized volatility window for SPX regime classification. Provides a stable signal while detecting monthly-scale stress periods.
RV Threshold	90th percentile	The top decile of SPX realized volatility days are classified as ‘stress’. Consistent with NBER recession-period concentration in the upper tail.

Note. All parameters are set before or during IS calibration. No parameter was adjusted after observing OOS results.

Appendix C: Synchronization Factor - Full-Series Statistics

Table C.1: Full-series summary statistics for the Synchronization Factor

Statistic	Value
Period	1999-07-09 to 2025-12-30
Total observations	6,661 trading days
k^* factors removed	1 (market factor)
Mean SF	1.8290
Median SF	1.7357

Std. deviation	0.4610
Minimum SF	1.0417
Maximum SF	4.4593
Days SF < 1 (below MP)	0 (0.0%)
Days SF > 1 (above MP)	6,661 (100.0%)
Days SF > 2 (high-risk regime)	1,890 (28.4%)
ACF (1)	0.959
ACF (30)	0.368
Half-life (ACF = 0.5)	≈17 trading days
Stress regimes mean SF	2.382 (667 days, 10.0%)
Calm regimes mean SF	1.767 (5,994 days, 90.0%)
Cohen's d (stress vs. calm)	1.455 (large effect)
Welch t-statistic	27.333
KS statistic	0.4704

Note. Regime classification: stress = SPX 30-day realized volatility > 90th percentile (27.44% annualized). Half-life is computed as the first lag at which the sample ACF crosses 0.5.